

MAT1100 Homework 1

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Question 5

(a) *Proof.* Given $f_{a,b}, f_{c,d} \in A$ and $t \in \mathbb{R}$

$$f_{a,b} \circ f_{c,d}(t) = f_{a,b}(ct + d) = a(ct + d) + b = act + ad + b = f_{ac,ad+b}(t)$$

and therefore

$$f_{a,b} \circ f_{c,d} = f_{ac,ad+b}$$

hence function composition is well defined as a binary operation on A . This is an associative operation as function composition is itself associative. Function composition has identity $I(t) = t$ for $t \in \mathbb{R}$ and this can be written as

$$f(t) = 1 \cdot t + 0 = f_{1,0}(t)$$

and $f_{1,0} \in A$ hence A has an identity for the group operation. Finally to find an inverse for $f_{a,b}$ consider arbitrary $f_{c,d}$ such that

$$f_{a,b} \circ f_{c,d} = f_{1,0}$$

applied to a real number t this becomes

$$f_{ac,ad+b}(t) = f_{1,0}(t)$$

which means

$$ac \cdot t + ad + b = t$$

now for $t = 0$ this implies that

$$ad + b = 0 \implies d = -\frac{b}{a}$$

which is well defined as $a \in \mathbb{R}^\times$. Then this implies that

$$ac \cdot t = t$$

for all real t and hence $c = \frac{1}{a}$. Therefore $f_{a,b}$ has right inverse $f_{\frac{1}{a}, -\frac{b}{a}}$ and to check this is a left inverse we compute:

$$f_{a,b} \circ f_{\frac{1}{a}, -\frac{b}{a}} = f_{a \cdot \frac{1}{a}, a \cdot (-\frac{b}{a}) + b} = f_{1, -b+b} = f_{1,0}$$

thus A is closed under left and right composition inverses and hence is a group with function composition as the group operation.

Now we define $A \curvearrowright \mathbb{R}$ by $f \cdot t = f(t)$. To show that this is an action fix $t \in \mathbb{R}$. Then

$$f_{1,0} \cdot t = f_{1,0}(t) = 1 \cdot t + 0 = t$$

and for $f_{a,b}, f_{c,d} \in A$

$$(f_{a,b} \circ f_{c,d}) \cdot t = f_{ac,ad+b}(t) = act + ad + b = a(ct + d) + b = f_{a,b} \cdot (ct + d) = f_{a,b} \cdot (f_{c,d} \cdot t)$$

hence the above is actually an action. Now to show that this action is transitive consider $s, t \in \mathbb{R}$ then take $f_{1,t-s} \in A$

$$f_{1,t-s} \cdot s = 1 \cdot s + t - s = t$$

hence the action is transitive. □

(b) *Proof.* First we show λ is a homomorphism. Given $f_{a,b}, f_{c,d} \in A$:

$$\lambda(f_{a,b} \circ f_{c,d}) = \lambda(f_{ac,ad+b}) = ac = \lambda(f_{a,b})\lambda(f_{c,d})$$

hence λ is a homomorphism.

Now to show λ is surjective. Suppose $a \in \mathbb{R}^\times$. Then

$$\lambda(f_{a,0}) = a$$

hence λ is surjective.

Finally I claim $\ker \lambda = T$. To show this, Note that $f_{a,b} \in \ker \lambda$ if and only if $\lambda(f_{a,b}) = 1$ which is equivalent to $a = 1$ and this is precisely what it means for $f_{a,b}$ to be an element of T . □

(c) *Proof.* That T is normal in A follows from the fact that it is the kernel of a homomorphism out of A . Now I claim that given $f_{a,b} \in A$,

$$f_{a,b}T = \{f_{a,x} : x \in \mathbb{R}\}$$

and indeed since

$$f_{a,b} \circ f_{c,d}^{-1}(t) = f_{a,b}\left(\frac{1}{c}t - \frac{d}{c}\right) = \frac{a}{c}t - \frac{ad}{c} + b$$

hence $f_{a,b}f_{c,d}^{-1} \in T$ if and only if $\frac{a}{c} = 1$ or $a = c$. which proves the above claim. Then to determine an explicit isomorphism from $\frac{A}{T} \cong \mathbb{R}$ we can simply consider λ restricted to the quotient. This follows as λ is constant on each coset of T in A and therefore

$$\lambda(f_{a,b}T) = a$$

is well defined, from part (b) it is clear that λ is still a surjective homomorphism. Then if $\lambda(f_{a,b}T) = 1$ this implies that $a = 1$ and hence $f_{a,b} \in T$ which is the identity coset for the group operation on A/T . Hence λ restricts to an isomorphism on the quotient □

- (d) *Proof.* Let $A \curvearrowright \mathbb{R}$ by evaluation. To determine A_0 note that $f_{a,b} \cdot 0 = 0$ if and only if $a \cdot 0 + b = 0$ which is equivalent to b being 0. Hence

$$A_0 = \{f_{a,0} : a \in \mathbb{R}^\times\}$$

The kernel of this action K is in particular a subgroup of A_0 and therefore if $f_{a,b} \in K$ then $b = 0$. Then we must have

$$f_{a,b} \cdot 1 = 1$$

which means $a \cdot 1 + b = a = 1$ and $K = \{f_{1,0}\}$ so this action is faithful.

However, A_0 is not a normal subgroup of A . To show this consider $f_{2,0} \in A_0$ and $f_{1,1}$ and $f_{1,1}^{-1} = f_{1,-1}$. Then we have

$$\begin{aligned} f_{1,1} \circ f_{2,0} \circ f_{1,1}^{-1}(t) &= f_{1,1} \circ f_{2,0}(1 \cdot t - 1) \\ &= f_{1,1}(2t - 2) \\ &= 1 \cdot (2t - 2) + 1 \\ &= 2t - 1 \end{aligned}$$

and this is not an element of A_0 hence it is not invariant under conjugation by elements in A and thus cannot be normal. $\square$

- (e) *Proof.* To determine $f_{a,b}A_0$ consider $f_{c,d}$ such that

$$f_{a,b} \circ f_{c,d}^{-1} = f_{\frac{a}{c}, b - a\frac{d}{c}} \in A_0$$

this holds precisely when

$$b - a\frac{d}{c} = 0$$

and therefore

$$\frac{b}{a} = \frac{d}{c}$$

fixing c this forces $d = c\frac{b}{a}$ and hence

$$f_{a,b}A_0 = \{f_{\alpha, \frac{b}{a}\alpha} : \alpha \in \mathbb{R}^\times\}$$

this determines a map:

$$\varphi(f_{a,b}A_0) = \frac{b}{a} \in \mathbb{R}$$

and by the above observation this is well defined. Furthermore given $c \in \mathbb{R}$

$$\varphi(f_{1,c}A_0) = \frac{c}{1} = c$$

hence φ is surjective and

$$\varphi(f_{a,b}A_0) = 0$$

implies $\frac{b}{a} = 0$ and hence $b = 0$. which means $f_{a,b} \in A_0$. Hence φ is injective. Thus, φ is a bijection.

To show that coset multiplication on A/T is well defined it suffices to show that given two cosets $f_{a,b}T$ and $f_{c,d}T$ given any element in the first coset and any element in the second,

their product will lie in the coset $(f_{a,b} \circ f_{c,d})T$. Concretely let $f_{a,b'} \in f_{a,b}T$ and $f_{c,d'} \in f_{c,d}T$ since all elements in the same coset of T have the same multiplicative element. That their product lies in $(f_{a,b} \circ f_{c,d})T$ follows from the fact that

$$f_{a,b'}f_{c,d'} = f_{ac,ad'+b'} \in f_{ac,ad+b}T$$

Now the same does not hold for $f_{a,b}A_0$ and $f_{c,d}A_0$ as if $f_{a',b'} \in f_{a,b}A_0$ and $f_{c',d'} \in f_{c,d}A_0$ this means

$$\frac{b'}{a'} = \frac{b}{a} \quad \frac{d'}{c'} = \frac{d}{c}$$

and we have

$$f_{a',b'} \circ f_{c',d'} = f_{a'c',a'd'+b'}$$

$$f_{a,b} \circ f_{c,d} = f_{ac,ad+b}$$

and for these to be in the same coset we would need

$$\frac{ad+b}{ac} = \frac{a'd'+b'}{a'c'}$$

and taking $a = b = a' = b' = 1$ and $d = d' = 0$ with $c = 1, c' = 2$ this would imply

$$\frac{1 \cdot 0 + 1}{1 \cdot 1} = 1 = \frac{1 \cdot 0 + 1}{1 \cdot 2} = \frac{1}{2}$$

which is clearly a contradiction, hence coset multiplication on A/A_0 is not well defined. $\square$